Bachelor of Science in Software Engineering
August 2026



Deficiencies and Usage Practices in Issue Tracking for Production Environments

Simon Långström

Marcus Pettersson

Faculty of Computing, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden

This thesis is submitted to the Faculty of Computing at Blekinge Institute of Technology in partial fulfillment of the requirements for the degree of Bachelor of Science in Software Engineering. The thesis is equivalent to 10 weeks of full-time studies.

Contact Information:
Author(s):
Simon Långström
E-mail: siln20@student.bth.se

Marcus Pettersson
E-mail: mapg23@student.bth.se

University advisor:
Dr. Usman Nasir
Department of Software Engineering

Faculty of Computing
Blekinge Institute of Technology
SE–371 79 Karlskrona, Sweden

Internet : www.bth.se
Phone : +46 455 38 50 00
Fax : +46 455 38 50 57

The authors declare that they are the sole authors of this thesis and that they have not used any sources other than those listed in the bibliography and identified as references. They further declare that they have not submitted this thesis at any other institution to obtain a degree.

Generative artificial intelligence (AI) has been used in this thesis as a supporting resource. The use has been limited to only supportive tasks. Tasks like these include: reviewing and improving grammar and sentence structure, assisting with data collection and pre-processing, and explaining concepts unfamiliar to the authors during research. It has not been used to replace or synthesize content, analysis, interpretation, or critical judgment.

Abstract

Background. Issue tracking systems are used in many different environments and contexts, yet there has not been much research regarding user experience within issue tracking systems in a production environment.

Objectives. This study aims to validate the issue-tracking system *Icheck* in a printed circuit board production environment, examine users' subjective thoughts and experiences, and determine how they relate to acceptance and continued usage.

Methods. The users' views were captured through a focus group. The questions were slightly rephrased versions of those from the Unified Theory of Acceptance and Use of Technology (UTAUT) questionnaire to better suit a group discussion. The findings consisted of qualitative data, which was then divided into themes to find similarities and patterns among the participants. This reduced the amount of data gathered without losing its meaning, while also improving the clarity and structure of the data. Themes were then used to draw conclusions of the research questions.

Results. The themes found were related to information and content quality, perceived efficiency, usability, and supporting infrastructure. Users reported that perceived usability and efficiency were low due to insufficient information, limited support, and missing content.

Conclusions. The findings suggest that this issue tracking system is perceived to have little positive impact on daily work performance, primarily due to usability challenges. The acceptance and continued use of the system are found to be dependent on effort expectancy from a user's perspective. It was also found that the system had experienced slowdowns and delays due to users' belief that the hardware was not powerful enough to run the system reliably. Overall, the findings for this study suggest that improving usability and further adapting the system to operator workflows can be used to increase efficiency, usability, and continued acceptance.

Keywords: Issue Tracking, Production environment, usage practices, UTAUT

Contents

Abstract ii

1 Introduction 1
- 1.1 Research Objectives . . . 1
- 1.2 Thesis Layout . . . 2

2 Background and related work 3
- 2.1 UTAUT . . . 3
- 2.2 Related Work . . . 5

3 Method 7
- 3.1 Study Design . . . 7
- 3.2 Empirical Study . . . 8
  - 3.2.1 Literature Search Procedure . . . 8
  - 3.2.2 Focus Group . . . 9
- 3.3 Data analysis procedure . . . 10
- 3.4 Method validity . . . 11

4 Results 12
- 4.1 Participants . . . 12
- 4.2 Codes . . . 12
- 4.3 Themes . . . 13
  - 4.3.1 Perceived Efficiency . . . 13
  - 4.3.2 Usability . . . 14
  - 4.3.3 Information and Content Quality . . . 14
  - 4.3.4 Supporting infrastructure . . . 15
  - 4.3.5 Summary . . . 16

5 Discussion 18
- 5.1 Answering RQ1 . . . 18
- 5.2 Answering RQ2 . . . 19
- 5.3 Answering RQ3 . . . 19
- 5.4 Limitations . . . 20
- 5.5 Practical Implications for PCB/PLSI Manufacturing . . . 20

6 Conclusions and Future Work 21

References 22

7 Appendix **25**
7.1 Questions 25

Chapter 1

Introduction

Issue tracking systems are tools used in software engineering to manage and resolve defects that occur during development. Grundy et al. [1] define issue-tracking systems as software tools used for internal bug tracking, enabling team members to report issues discovered during product testing to the development team. Typically, issues are submitted as tickets, which allow problems to be documented, assigned, and tracked until they are resolved.

Moreover, the general problem-tracking industry is projected to grow to 23.7 billion USD in 6 years and is increasingly adopted across industries [2]. Several companies already provide comprehensive issue-tracking tools for hardware production environments [3, 4], indicating that this field is highly relevant and continues to grow. Previous research has investigated how personalized interfaces in issue tracking systems can improve situational awareness [5]. Similar to software engineering, hardware manufacturing requires reliable, structured methods for reporting and resolving errors that arise during the production process. In this case, an issue-tracking system can be used to manage production-related problems.

By conducting this thesis, we expect to identify and explore the causes of why issue tracking systems are not as efficient as they could be, especially in a factory setting, by investigating operators view of the issue tracking system *Icheck*. This generates practical value for the organizations in which the study is conducted by informing existing workflows and usage practices. A previous study found that a more improved and interactive environment for reporting software bugs was highly perceived as useful [6].

1.1 Research Objectives

The following research questions were specifically chosen to provide insight into the user experience and the technological acceptance of issue-tracking systems. Furthermore, these questions were aimed to explore users' perception of the system rather than its actual effect, since this thesis aims to answer questions regarding user experience, which is how individuals perceive the system.

- **RQ1:** How do operators in a PCB/PLSI (Printed Circuit Board / Programmable Large Scale Integration) production environment perceive the impact of an issue tracking system on their daily work performance?
- **RQ2:** How does the overall user experience affect perceived efficiency and user satisfaction in issue tracking systems in a PCB/PLSI production environment?

- **RQ3:** What factors influence the acceptance and continued use of an issue tracking system in a PCB/PLSI production environment?

1.2 Thesis Layout

The study aims to contribute knowledge on usage factors that limit the effectiveness of issue tracking systems, as well as how user experience influences their perceived effectiveness and acceptance. Earlier studies have examined the user experience and technology acceptance of various tools in industrial settings; however, based on the literature review for this study, no research was identified that addresses these areas within issue-tracking systems in industrial environments.

Chapter 2

Background and related work

This chapter will introduce the theoretical foundation and its key concepts upon which this thesis is built. A literature review was also conducted to provide a general overview of the current state of research in the area of study. The approach used is shown in chapter 3. The Unified Theory of Acceptance and Use of Technology (UTAUT) serves as the framework for this thesis. The core determinants of the framework are then described to establish their meaning. The chapter also discusses the predecessor TAM, what it is, and how it is used. Later in the chapter, previous work is introduced, exploring studies that have examined usability and workflows in issue-tracking systems.

2.1 UTAUT

The Technology Acceptance Model (TAM) was proposed by Fred D. Davis in 1989 as a common model used to explain and anticipate user acceptance [7]. The model has been used to evaluate a wide range of applications in various environments, such as educational technology, e-banking, and other types of software [8]. It was designed to gauge the acceptance and use of technological systems. TAM proposes two different key factors that influence technology acceptance: perceived usefulness and perceived ease of use. *Perceived usefulness* relates to the extent a user perceives a system improving their performance, while *Perceived ease of use* relates to the extent a user perceives the system being effortless [9]. In 2003, Venkatesh et al. developed the Unified Theory of Acceptance and the Use of Technology (UTAUT) as an advanced extension of TAM, integrating it with several other technology acceptance models, such as the Theory of Reasoned Action, into a unified framework [9].

UTAUT explains how individuals form intentions to use a technological system and how these intentions translate into actual system use. The model proposes four key determinants of technology acceptance: performance expectancy, effort expectancy, social influence, and facilitating conditions. Rather than considering these determinants independently, UTAUT describes how they contribute to behavioral intention and use behavior.

Figure 2.1 illustrates an overview of the model that describes how the determinants interact and influence the intentions of behavior and usage, adapted and redrawn from Venkatesh et al.'s paper [9] to highlight the relevant parts we will examine in the study. The red color for the moderating conditions in the bottom of the diagram is simply for better readability, since they influence all four other determinants in the model. The determinants are as follows:

- Social Influence: Social influence is defined as the degree to which an individual perceives that important others believe he or she should use the new system [9, p. 451]. In an organizational environment, this could include colleagues, supervisors, or other members of the organization.

- Effort Expectancy: The degree of effort that is expected to be used when working with the system [9, p. 450]. In the context of an issue-tracking system, this can relate to how easily users can navigate the system, create and update reported bugs, find relevant information, and understand the different functions within the system without requiring extensive training or assistance.

- Performance Expectancy: Performance expectancy is defined as the degree to which an individual believes that using the system will help him or her to attain gains in job performance [9, p. 447]. In the context of an issue-tracking system, this can relate to whether users perceive the system as helping them manage, communicate, and resolve production-related issues more effectively.

- Facilitating Conditions: Facilitating conditions are defined as the degree to which an individual believes that an organizational and technical infrastructure exists to support use of the system [9, p. 453]. In the context of an issue-tracking system, this can relate to whether users have access to the necessary hardware, software, training, documentation, and technical support required to use the system effectively. It can also include whether established organizational processes support the use of the system for reporting and managing production-related issues.

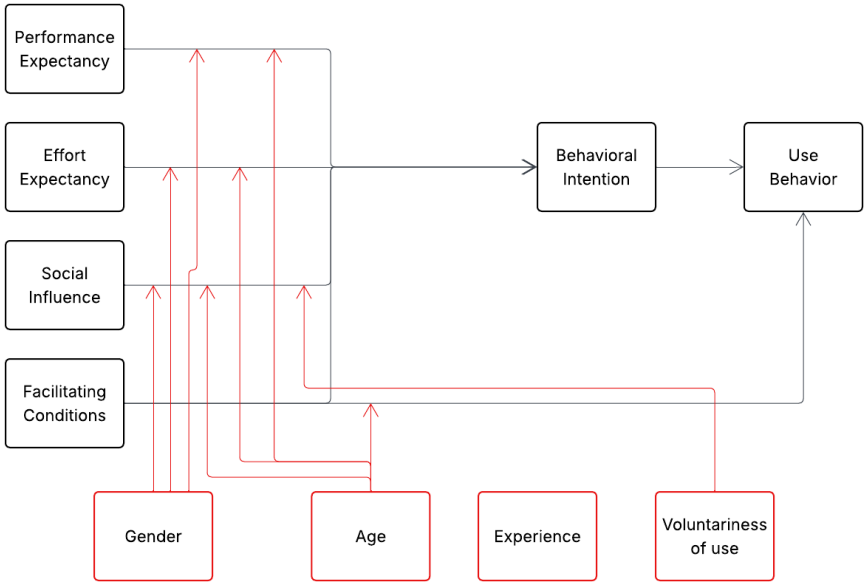


Figure 2.1: A map of the UTAUT model, with the moderating determinants in red.

Within UTAUT, the four determinants are related to two main outcomes, behavioral intention and use behavior. Performance expectancy, effort expectancy, and

social influence contribute to behavioral intention, which in turn influences actual use of the system. Facilitating conditions have a direct influence on use behavior, as the availability of appropriate organizational and technical resources can enable or constrain a users' ability to use the system. In this way, UTAUT provides a framework that helps understand why users intend to use the system and the conditions that support its actual use.

In the context of this study, the model can therefore be used to examine how users' perceptions of the issue-tracking system relate to their acceptance and use of the system. An example of this is a user who perceives that the system improves their work performance could also have a stronger intention to use it, and if a user finds the system difficult to use, the user may be less inclined to use it. Similar encouragement from colleagues or supervisors may also influence users' willingness to use the system, while sufficient technical and organizational support may facilitate its actual use. These relationships provide the theoretical basis for applying UTAUT to the issue-tracking system investigated in this study.

2.2 Related Work

TAM has previously been used to study the use of ERPs (Enterprise Resource Planning Systems) in various ways. Sternad and Bobek [10] proposed an extension of the TAM model, consistent with their findings, which suggested that work compatibility and system characteristics influence perceived usefulness and ease of use. Another paper by Jo and Park [11] analyzed the mechanisms that support ERP management from the perspectives of system quality and user information processing. Using the TAM model, different hypotheses were formulated to identify the mechanisms. It developed a model that showed how the mechanisms influenced the satisfaction and perceived usefulness of an ERP, and linked the user experience to perceived ease of use and system reliability.

Singh [12] has explored the design flow and general usability of an issue-tracking system for a silicon chip manufacturer before, through a case study that quantitatively examined the workflow and usability. Here, it was found that the usability of a tool in that environment was defined by its successful usage and that the tool helped users improve. The case study was conducted in a similar setting to that of our thesis, although the manufacturing environment differed because it involved silicon-based chips rather than PCB/PLSIs. While the production processes differ between PCB/PLSIs and silicon-based chips, both need reliable systems for issue and defect tracking. For example, Liu et al. [13] created a tool with a graphical interface for detecting root causes of these defects. The design process was analyzed in depth, together with a quantitative analysis of production and usability by mapping goal-question-metrics to measurable metrics.

An earlier work-in-progress paper by Lessel et al. [14] presented a prototype of such a system but did not report results from an empirical user study. In contrast, our work is a study that evaluates an existing system using theories of user experience evaluation and technology acceptance. The system evaluated in our proposed work employs a different design; however, the findings are still expected to provide valuable insights into best practices and operator behavior in issue-tracking systems on the

shop floor. Regarding actual user experience evaluations, prior research has examined survey-based methods and context-specific metrics, but not the area of focus of this study [15, 16]. Innecco et al. [17] had examined the usability and acceptance of a Manufacturing Execution System with survey questions by quantitatively comparing responses from multiple groups, and found that *Facilitating Conditions* and *Effort Expectancy*, the latter for young users, influence *Usage Behavior* and *Behavioral Intention*, all based on the UTAUT theory of [9]. The change in methodology and scope from the study in this thesis allows us to examine perceived usability and acceptance in a deeper, narrower context.

Overall, the findings from these studies indicate that the experience of using a technological system is influenced by several interconnected factors. Sternad and Bobek [10] and Jo and Park [11] both highlight the importance of system-related factors in shaping users' perceptions of usefulness and ease of use, although they approach these factors from different perspectives. This is also reflected in the work of Singh [12], where usability was examined specifically in relation to an issue-tracking system used in a manufacturing environment. While Singh [12] focuses more directly on usability and workflow than technology acceptance, its findings complement the results of Sternad and Bobek [10] and Jo and Park [11] by demonstrating the importance of how a system supports users in their work. Similarly, Innecco et al. [17] identify effort expectancy and facilitating conditions as factors related to users' intention and use of a manufacturing system, further indicating that both the system itself and the conditions surrounding its use can influence users' interaction with technology. Taken together, these studies suggest that usability, system characteristics, and the surrounding work environment are important considerations when evaluating users' experiences with production-related systems. However, the studies differ in how these factors are evaluated and in the contexts in which they are investigated. This makes it difficult to determine how these factors interact specifically within issue-tracking systems used in PCB/PLSI production environments, which are the focus of the present study.

Chapter 3

Method

This thesis examines the issue-tracking system as a smaller subsection of the Manufacturing Execution System from the perspective of a smaller, more isolated group of end users. Previous studies have primarily focused on larger systems in environments similar to our environment. However, few studies have focused on the perceived user experience within specific subsystems and single user groups. Furthermore, much of the previous research has examined Enterprise Resource Planning Systems. The issue-tracking system examined in our study could be considered a subsystem of a Manufacturing Execution System, operating at a more specialized level, although it could still be considered similar to an ERP, given the organizational context it serves.

3.1 Study Design

To gather the data needed for this study, a qualitative approach was designed and adopted, with focus groups as the chosen data collection method. This study favored qualitative research methods, such as focus groups, due to a shortage of participants, as these are generally better suited to studies with limited data. Furthermore, when addressing the research questions for this thesis, a quantitative approach was difficult to use for questions about individual perspectives, as it requires reducing phenomena to numerical values and focuses on measurable associations [18].

The figure below shows the process used in the study design. We conducted a focus group to collect end users' views on *Icheck* (the issue-tracking system). The focus group transcript was used to conduct the thematic analysis. This transcript was then systematically coded by identifying recurring keywords within the responses. These codes were then organized and sorted into UTAUT's core determinants. These findings, along with those from the literature review, were used to answer our research questions.

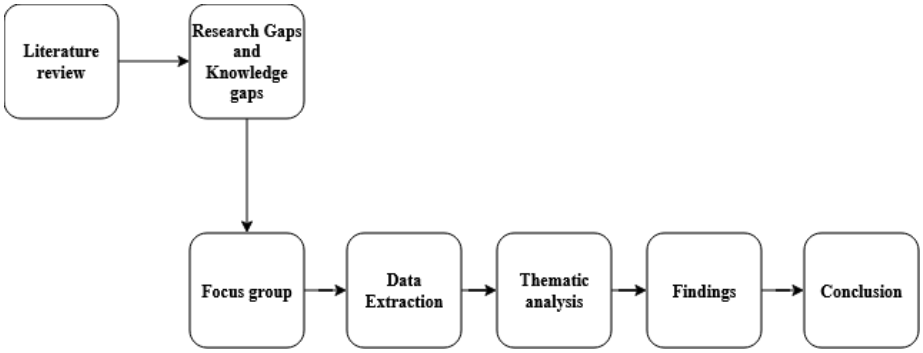


Figure 3.1: Process of study design

A qualitative approach is better suited to measuring individual perspectives, making it well-suited to addressing the research questions. Moreover, a mixed-methods research design would not be suitable for this type of thesis, as it would exceed the time allotted. Using the mixed approach as a research method could be suitable for this type of thesis if the scope is clearly limited and the design is focused from the start.

- Interviews as a research method could be a suitable option, as they are a qualitative method that allows for in-depth exploration of the subject while removing the possibility of group bias, which focus groups risk introducing. However, there is a risk that an interview can focus too much on the individual perspective, limiting exploration of how user perceptions are shaped, challenged, or reinforced in social interactions. A guide written by the Illinois Institute of Technology Library that evaluates the different research methods also states that interviews often result in "Limited ability to capture context, meaning, and subjective experiences" [19].

- Using a survey was considered, but since it is a quantitative method, it makes it difficult to measure individual perspectives. Conducting one also requires many more participants to yield useful data, a resource this thesis lacked.

Since the empirical part of the thesis included a total of five subjects, a qualitative approach was most appropriate, with a focus group well-suited to address the research questions. Other field-based methods were considered but ultimately ruled out due to the study's constraints, chiefly the inability to access users during real-world use of the system.

3.2 Empirical Study

3.2.1 Literature Search Procedure

A literature review was conducted to establish the theoretical foundation of the study, identify research and knowledge gaps, and support the analytical framework. To identify relevant literature, the referenced works were limited to scientific, peer-reviewed, and academic sources in a semi-structured, non-systematic manner. While

non-systematic, this method allowed sources to be identified with the broader area of interest without a longer, more time-consuming review, which still would generate a broader overview of the current state of research in the subject would be largely irrelevant to answer our RQs for the study. Works without references were to be excluded.

IEEE Xplore, University libraries, Google Scholar, and other scientific sources were used to search for relevant literature, and the selected works were assigned keywords after final selection. The preliminary search query used in the literature search was defined as:

> ("defect tracking" OR "issue tracking" OR "nonconformance" OR "quality management system" OR QMS OR MES) AND ("usability" OR "user experience" OR "perceived efficiency" OR UTAUT OR TAM OR "ease of use") AND ("manufacturing" OR "production" OR "industrial" OR "assembly" OR "electronics manufacturing").

For the selected works, a backward snowball approach was also used, reviewing publications that cite them and including relevant studies, according to the guidelines provided in [20]. In addition, works referenced in related or future research sections were to be considered for inclusion. The literature was then analyzed to identify its key themes and findings relevant to the research questions, including operational adoption and perceived usability of error-tracking systems. The inclusion of the papers was determined by whether they contained issue tracking, nonconformance reporting, incident reporting systems, usability, accessibility, or organizational learning in manufacturing or industrial environments. After this, a final selection of literature was made based on the previously established criteria.

An earlier systematic literature review conducted by Nur [21] identified three primary categories of user experience evaluation methods: observational, self-reported, and physiological, with self-reported methods the most commonly used. The review also noted that two papers evaluated user experience in industrial settings, none of which addressed issue-tracking systems. For our study, the research questions focus on users' subjective experiences, beliefs, and attitudes, requiring methods that can reliably capture these aspects. The research questions of this study primarily concern perceived user experience, which suggests that physiological methods would be of limited applicability. These methods examine bodily responses and user experiences during system interaction by using indicators such as heart rate and eye movement, and are therefore appropriate for evaluating stress and bodily reactions, metrics that are not examined in this study. Observational methods, along with performance and behavioral metrics, could complement the self-reporting method used in this study to validate and reinforce certain user statements, but are otherwise insufficient to capture the study's core focus.

3.2.2 Focus Group

The empirical study consisted of a single focus group. The focus group session was conducted with technicians who work with the *Icheck* issue-tracking system on a day-to-day basis and with team leaders. A total of 5 participants attended the focus group session voluntarily. The discussions were recorded with their consent, and

additional notes were taken to simplify the collection of data. The ages of each participant were also collected.

To ensure active, balanced participation and prevent individuals from dominating the discussions, we actively managed turn-taking by directly inviting input from quieter participants, such as asking for agreement, disagreement or alternative experiences. We also had a lot of follow-up questions which ensured that all participants contributed across all the questions. We as moderators remained completely neutral and only took notes and guided the conversation when needed.

The focus group was asked questions related to the research questions, and the generated data was used to answer them and support the findings of this thesis. The questionnaire was derived from the UTAUT model and therefore focuses on several factors, including perceived efficiency, user satisfaction, user experience, and operational acceptance. The set of questions used in the study is shared in section 7.1. Questions in italics correspond to the items in the UTAUT questionnaire, while the bold questions following them are the questions asked in the discussion based on these. In this sense, UTAUT is used as the theoretical framework for the focus group questions, but the original questionnaire is not used.

These were used to provide reliable answers to previously posed research questions by qualitatively analyzing data from the focus groups.

The data collected from the focus groups was then analyzed using a qualitative thematic analysis [22]. It involves identifying and reporting patterns within the data, which are then presented textually. A detailed description of the process is in section 3.3

Data was additionally recorded on audio for transcription. To transcribe the recording, the tool noScribe was used [23]. This tool uses AI to transcribe audio to text. The transcriptions were also manually processed to address errors and inaccuracies introduced by the AI. The transcript was then coded by analyzing the various patterns and assigning codes. For instance, the statement *"And then it takes time away from production"* was coded as a workflow disruption. This indicates that the participant perceives that the technology is interfering with existing work processes.

3.3 Data analysis procedure

Braun et al. [22] describe six phases of data analysis, which were followed in order to get the codes and themes. The first step details the process of reading and examining the data itself, which was described in the previous section, e.g., reading the transcription and listening to the audio.

For the second step, involving the generation of the initial codes, the transcript was systematically examined by assigning codes to the data based on identified meanings and patterns. Color-coded markings were then used to distinguish the relevant sections of the transcript.

As the themes had been determined beforehand, the third, fourth, and fifth steps were conducted by grouping each code according to its corresponding theme. In cases of disagreement, a conclusion was reached either through discussion and compromise or through a final decision by one of the researchers. Some overlap between themes

was permitted; however, care was taken to avoid such overlap for the most frequently occurring codes.

3.4 Method validity

To collect qualitative data on participants' perceptions, a voluntary focus group was held, and invitations were sent to each production floor department coordinator. The focus group consisted of five participants, and the session followed a semi-structured format, with open-ended questions guiding the discussion. Participants were allowed to freely elaborate on and discuss the questions with one another. By doing a focus group, multiple perspectives were gathered in a limited time, which was necessary given the small number of participants and the limited time frame in which the thesis was written.

The focus group format allows participants to comment on and respond to statements. This, in turn, generates data for how each of their statements are expressed, questioned, or supported in a social context. However, this may also lead some participants to speak more than others, and, in turn, may have influenced certain viewpoints. We discussed this more in-depth in section 3.2 and how we came to mitigate it. As participation in the group was voluntary, there is a risk of a self-selection bias [24], leading to the possibility that participants who chose to participate may have had a greater interest in the topic or stronger opinions than those who did not.

Clarifying questions were asked to ensure the focus group's validity and reduce the risk of misinterpretation. The discussion was recorded and supported with notes for easier data review. Reliability is limited by a single focus group, so results cannot be compared across sessions to assess both consistency and inconsistency. Instead, reliability was attempted by using a semi-structured format with open questions based on the UTAUT framework. This was also supported by recording and taking notes during the session, but since the findings depend on the dynamics of a specific group, they should be interpreted with caution.

The questions were constructed by adapting the UTAUT questionnaire created by Venkatesh [9] to better suit open discussion.

Chapter 4

Results

This chapter discloses the findings of the focus group. The codes that have been extracted are separated and sorted into themes. All of the codes and their frequencies can be found under section 4.2. The themes and their respective codes can be found in tables under their respective theme in section 4.3.

4.1 Participants

There were five participants in the focus group. Their ages ranged from 37 to 41 years, and they worked on different assembly teams. Two participants also worked with QA in parallel with their assembly roles. This additional QA experience may have influenced their perspectives on the issue-tracking system, as they were familiar with the system from both an assembly and a quality-assurance perspective. Their responses may therefore reflect a broader understanding of the system than those of participants working solely in assembly. However, the focus group data did not allow us to clearly distinguish whether differences in responses were attributable to their QA experience.

4.2 Codes

All codes are listed in section 4.2, and are then grouped into themes in section 4.3. In order for UTAUT to be applicable, the themes were chosen at the start to be the determinants in UTAUT that this thesis explores. Information and Content Quality and Supporting infrastructure are connected to *Facilitating Conditions* and *Social Influence* since they pertain to technical and organizational infrastructure. Meanwhile, Perceived Efficiency and Usability pertain to *Performance Expectancy* and *Effort Expectancy* respectively. The codes were derived from data that was then grouped into the already established themes.

Code	Frequency of appearance
Disruption of workflow	7
Insufficient support	7
No feedback on reported errors	5
Insufficient information	4
Technicians have not supplied correct content	4
Feeling of Confusion during use	3
Lack of system reliability	3
Lack of usability	3
Content is missing	3
BOMs / items not updated	3
Missing drawings	3
Missing images	3
Voluntariness	3
Acceptance	2
Effort Expectancy	2
Inconsistent hardware	2
Insufficient Knowledge	2
Lack of Perceived Efficiency	2
Age	1
Confusing user elements	1
Difficulty to find previous errors	1
Lack of features	1
Lack of information	1
Lack of specificity in system	1
Inconsistent information	1
Inconsistent Updates	1
Inconsistent Usage	1
Older items are worse	1
Varying quality of content	1

Table 4.1: All of the extracted codes together with the frequency they appeared in the discussion.

4.3 Themes

4.3.1 Perceived Efficiency

Perceived efficiency was a recurring theme in our discussions, shown in subsection 4.3.1. Participants identified several problems with the system that affected its efficiency. The reported consequences of these issues varied, though they were most commonly described as disrupting their workflow and, at times, directly linked to perceived inefficiency.

> *Sometimes it feels like it takes time to go to the data, and I mean, notice what's wrong than just working and getting on with it.*

The statement highlights core tension between system usage and shop-floor productivity, while the phrasing "than just working and getting on with it" shows that operators perceive interactions with the system as an administrative burden rather than being a useful support tool.

> *Sometimes it is pretty useful, but it glitches out (sic) often, and blueprints and other things don't catch up. And then I end up not using it.*

This statement highlights the direct link between system reliability and user adoption. While the participants acknowledge that the system has the potential to be "pretty useful," this shows how technical bugs directly cause workers to stop using the system, as stated in the quote "And then I end up not using it". This turns minor frustration into abandonment of the system, resulting in low perceived efficiency.

Code	Frequency of appearance
Disruption of workflow	7
Lack of Perceived Efficiency	2

Table 4.2: The codes that have been associated with Perceived Efficiency.

4.3.2 Usability

Participants frequently reported difficulties with system use. Inconsistencies in the presentation of information and system functionality would lead to confusion. They agreed that a common contributing factor to these problems was insufficient support when interfacing with the system. This was the code that appeared most frequently in the discussion on usability in subsection 4.3.2. Moreover, they also expressed that it was not always clear how to resolve problems within the system itself, perceiving this lack of guidance as discouraging the use of the system.

Code	Frequency of appearance
Insufficient support	7
Feeling of Confusion during use	3
Lack of system reliability	3
Lack of usability	3
Lack of information	1
Lack of features	1
Lack of specificity in system	1
Confusing user elements	1

Table 4.3: The codes that have been associated with usability.

4.3.3 Information and Content Quality

All participants agreed that the content quality within the system was unreliable and often varied across the products it contained. Many reported that missing blueprints

were common, making it difficult for them to perform their tasks. In addition, insufficient information and instances of missing content were discussed.

> *... Sometimes you only have a ground blueprint with directions and not much other information. We can ask the technicians to add an image of this, so that it looks like this and not like that (sic), or additional information. But it depends on how persistent you are yourself and how willing the technicians are to include it.*

It was also reported that the quality of content varied between the products. Some were documented, while others lacked important information. Inconsistencies were stronger for older products, which were associated with reduced usability and reliability among participants.

> Participant 1: *Yes, it is not consistent across all programs. It is not available for all articles. Some are well made, while others are not made at all.*
> Participant 2: *But it is probably, one could say, worse for older articles, but...*

The participants often have to follow up with the technicians that maintain the content of the software to ask questions and ask them to re-do parts of it.

> Pretty often (sic) we have to go to the technicians to ask questions and ask them to redo the blueprints and other stuff to make the system work.

The participants also discussed error handling within the software and noted that it was difficult to find previous errors for specific articles.

Code	Frequency of appearance
Insufficient information	4
Technicians have not supplied correct content	4
Missing drawings	3
Missing images	3
Content is missing	3
BOMs / items not updated	3
Varying quality of content	1
Older items are worse	1
Difficult to find previous errors	1

Table 4.4: The codes and frequencies related to Information and Content Quality.

4.3.4 Supporting infrastructure

While comments on the supporting organizational and technical infrastructure for the system were fewer than for some of the other themes, there were still insights to be drawn. A key issue was a perceived lack of feedback and response from the wider organization, alongside uneven encouragement of system use and general usage within teams. Participants also suggested that lower usage may be influenced by age and general lack of interest, as well as perceptions that the required effort was not worth the expected benefit.

Participant 1: *Someone never comes out and asks what happened, or examines why. And there can be a soldering right at that place for five years.*

Participant 2: *Yes, we never get any information back based on what we entered into the system.*

Code	Frequency of appearance
No feedback on reported errors	5
Voluntariness	3
Inconsistent hardware	2
Acceptance	2
Effort expectancy	2
Insufficient knowledge	2
Poor information structure	1
Age	1
Inconsistent updates	1
Inconsistent information	1
Inconsistent usage	1

Table 4.5: The codes and frequencies for Supporting Infrastructure.

4.3.5 Summary

The most prominent codes were *Disruption of workflow* and *Insufficient support*, while there was minimal discussion around others, such as inconsistencies in information, updates, and usage. An unexpected observation was the relatively high frequency of codes focused on technical infrastructure. This could indicate that it may play a larger role in continued usage and acceptance than was initially thought, especially in a production environment.

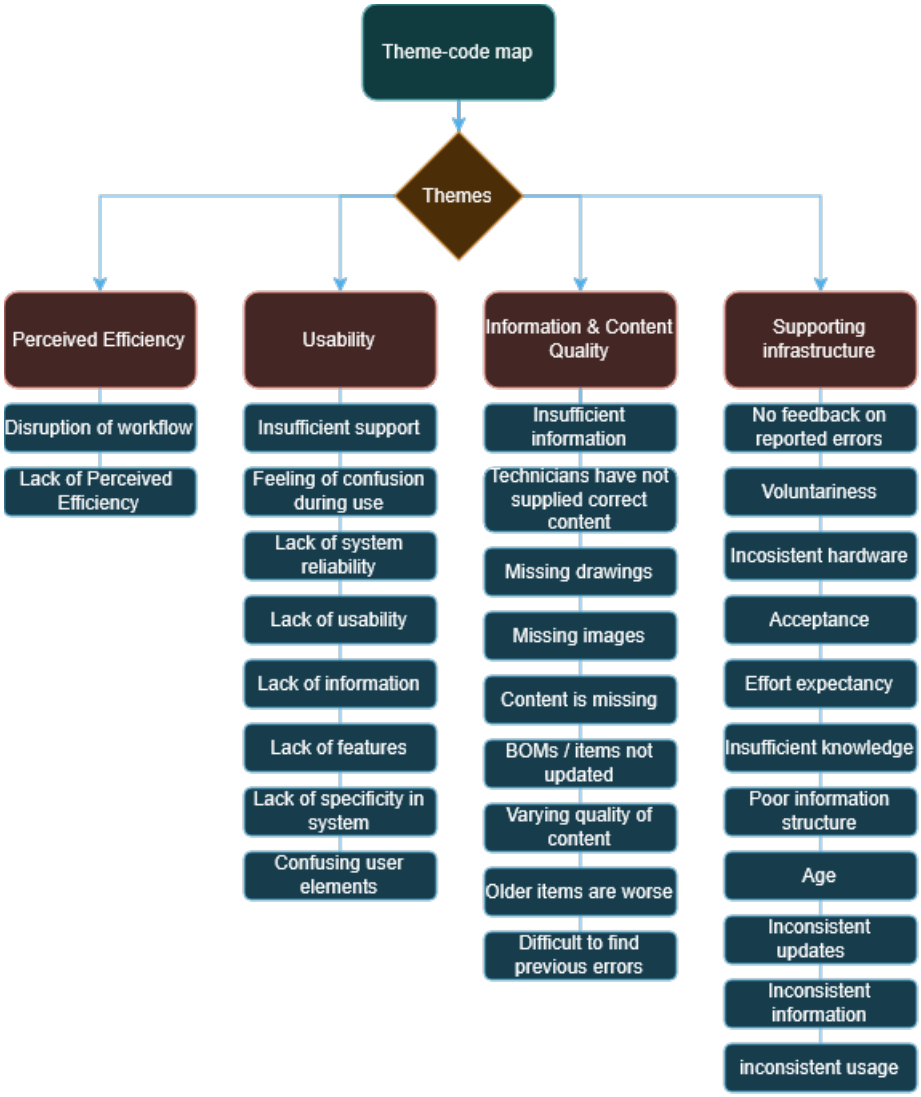


Figure 4.1: Map of all themes and their codes

Chapter 5

Discussion

This chapter discusses the empirical study's results in relation to the research questions. This chapter explains what the findings mean and how they answer the research questions. The results are also reflected upon and put into a broader context. Additionally, the limitations and implications of this thesis are discussed and reflected upon alongside the results.

5.1 Answering RQ1

The first research question explores how operators perceive the system's impact on their daily work performance in a PCB/PLSI environment, as in our study. The findings are reflected mainly in performance expectancy and effort expectancy, according to the UTAUT model [9]. The performance expectancy was low, since the participants did not perceive the system as enhancing their ability to complete tasks more efficiently or effectively. This is reflected in participants' descriptions of disrupted workflows, where logging issues were viewed as an interruption rather than a support mechanism. One operator highlighted this friction directly:

> Sometimes it feels like it takes time to go to the data, and I mean, notice what's wrong than just working and getting on with it

This sentiment illustrates how high perceived effort directly depresses performance expectancy. When operators feel that interacting with the system delays their primary assembly tasks, the perceived value of the tool diminishes. According to UTAUT, higher perceived effort paired with low performance gains leads to reduced behavioral intention to use the system [9]. Consequently, unrecorded defects remain in the workflow, leading to further process bottlenecks downstream.

The effort expectancy is also negatively affected, as usability issues, insufficient support, a confusing interface, and the system's reliability increase the effort required to use the system efficiently. As stated by [9], effort expectancy is a key determinant of behavioral intention, where higher perceived effort is associated with lower behavioral intention. Moreover, issues related to the system software, such as defects or reduced use of the issue-tracking system, may result in fewer issues being properly documented. This can slow down the overall process, as noted by [25].

5.2 Answering RQ2

Concerning RQ2, which explores user satisfaction and its effect on perceived efficiency and participant satisfaction, the findings of the study indicate that end-user satisfaction is strongly influenced by system constraints and user experience challenges. The results were consistent with prior research suggesting that ease of use and usability are key factors in user satisfaction [26], which, in turn, affects users' overall assessments of the system's effectiveness. However, the presence of relevant themes in the focus group does not necessarily imply that the broader user population experienced lower satisfaction. This is due to both moderating factors in the UTAUT model, such as age and gender, and to the general limitations of qualitative studies. To conclusively determine whether satisfaction decreased alongside perceived efficiency in this study, more in-depth research is required, such as a larger-scale quantitative or qualitative exploration using multiple evaluation methods.

5.3 Answering RQ3

The third research question in this study concerned the factors influencing the acceptance and continued use of an issue-tracking system in a PCB/PLSI production environment. The findings indicate that several factors contribute to both acceptance and continued use, consistent with previous research of the UTAUT model [9]. In UTAUT, effort expectancy is identified as a determinant with a direct effect on behavioral intention. In this study, several themes emerged from the focus group that align with the concept of Effort Expectancy, such as those related to the perceived difficulty of using the system. These findings, in the context of UTAUT, can indicate that effort expectancy is an important factor influencing behavioral intention, which, in turn, influences the acceptance and continued use of the system in the study. The remaining themes can be interpreted similarly in relation to the UTAUT determinants, although some findings, particularly those related to hardware constraints, were not fully captured by the model. Consequently, the emerging themes of the focus group can be understood to influence the continued use and acceptance of the system for this study design, as per the model.

However, because the analysis was conducted using codes that were organized to the UTAUT categories, the alignment between the identified themes and the model's determinants was somewhat expected. While the findings therefore support the applicability of UTAUT in this context, the study also identified factors that extend beyond the determinants explicitly described by the model. An example of this is the hardware-related issue that emerged as an important factor influencing users' experiences with the system. As a result, in this type of environment, acceptance and continued use may also be influenced by practical constraints related to supporting technical infrastructure.

In this study, the moderating variables of age, experience, gender, and voluntariness of use were not treated as core factors, despite their role as moderators in UTAUT. While these factors are acknowledged as important in prior research [9], the qualitative nature of this study and the limited number of participants limit the ability to draw meaningful conclusions based on participant characteristics. Despite

the group's diversity, we still believe conclusions can be drawn. Further discussion of these limitations is provided in section 5.4.

5.4 Limitations

The study in this thesis had limitations in its data collection phase, as the focus group comprised only five participants. This could affect the variety of perspectives and limit the extent to which the findings can be generalized. However, the focus group included participants working in different parts of the production, which, in turn, might have made them diverse enough to capture different viewpoints reliably despite the small group. Although limited, we believe the results can address the research questions within the scope of this study. It was conducted on a small subset of the overall population using the system, and the focus group was held only once, with limited opportunities to ask further questions after the discussion. Voluntariness also introduces the risk of self-selection bias, potentially leading to participants with stronger opinions being overrepresented and to inconsistent emphasis on some themes.

5.5 Practical Implications for PCB/PLSI Manufacturing

The results and answers to the research questions support both the UTAUT and TAM models and indicate that perceived efficiency and usability are significant factors for users in industrial PCB/PLSI environments. Models proposed by Venkatesh et al. [9, 27] suggest that these factors contribute to the decrease in acceptance and use of such a system. A separate, independent variable identified in the focus group was that the hardware on which the system ran was not always able to support it reliably, resulting in slowdowns and lag. Even delays of more than one second can make users feel as though they are waiting for the system to respond, while delays exceeding ten seconds may lead to a loss of attention to the task [28]. This study did not have any metric-based data collection, but even simply perceived slowdowns can negatively affect users' perception of ease of use, increasing the perceived difficulty of completing tasks effectively by interrupting their attention, as exemplified by this quote. In line with UTAUT, this can negatively influence system usage, along with the other themes identified in this study. Reduced usage can lead to fewer reported errors, which in turn may require resources to be relocated later to address issues that were not previously identified. In contrast, successfully reported errors could have led to earlier mitigation on boards of the same model. In the context of this study, unreported issues may recur in subsequent production cycles, necessitating their identification and repair. This results in additional work and increased resource consumption, although this is not shown in this study by itself. In particular, repeated repairs may involve additional soldering, which has been shown to have negative environmental impacts for both lead-based and lead-free alternatives [29].

Chapter 6

Conclusions and Future Work

The perceived impact of issue-tracking systems, user experience, and factors influencing continued use are difficult to evaluate based on a single focus group. However, the data do provide valuable insights. As a qualitative study based on a single focus group, these findings should be interpreted as exploratory rather than generalizable. The results of this study suggest that, in this specific context, all three research questions are closely related to human-computer interaction, user experience, and user satisfaction. In particular, the perceived efficiency appears to depend on factors such as system reliability and limitations in the supporting hardware on which the system is deployed, when interpreted through existing theoretical models. As discussed in section 2.2, prior research has quantitatively examined factors influencing system acceptance in similar environments [17]. While the findings of the present study correspond to established determinants in models such as UTAUT, participants also highlighted issues related to slow or insufficient hardware performance. This factor is not explicitly accounted for in either UTAUT or TAM. Given its frequency in the focus group, it is necessary to further investigate the relationship between hardware performance and user interaction in manufacturing environments where hardware conditions vary across workstations. More comprehensive studies should be conducted to capture perspectives not represented in this study to provide a more complete understanding of the themes identified among system users. Future research should investigate this relationship through both qualitative and quantitative approaches. In particular, studies that statistically examine the link between hardware performance, perceived efficiency, and user satisfaction could help clarify its impact and explore how this factor may interact with determinants and constructs in the UTAUT model.

References

[1] J. Grundy, H. Khalajzadeh, J. McIntosh, T. Kanij, and I. Mueller, "Evaluation of novel approaches to software engineering," in *Evaluation of Novel Approaches to Software Engineering: 15th International Conference, ENASE 2020, Prague, Czech Republic, May 5–6, 2020, Revised Selected Papers*, vol. 1291 of *Communications in Computer and Information Science*, pp. 444–468, Springer, 2021.

[2] T. Kumbhar, "Problem tracking software market to reach usd 23.66 billion by 2032," tech. rep., Introspective Market Research, July 2024.

[3] Issuetrak, "Manufacturing." `https://www.issuetrak.com/solutions/manufacturing`. Accessed: 2026-02-24.

[4] Eviview, "Manufacturing issue tracking." `https://www.eviview.com/rca/manufacturing-issue-tracking/`. Accessed: 2026-02-24.

[5] O. Baysal, R. Holmes, and M. W. Godfrey, "Situational awareness: Personalizing issue tracking systems," in *Proceedings of the 35th International Conference on Software Engineering (ICSE)*, pp. 1185–1188, 2013.

[6] Y. Song, J. Mahmud, Y. Zhou, O. Chaparro, K. Moran, A. Marcus, and D. Poshyvanyk, "Toward interactive bug reporting for (android app) end-users," in *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, ESEC/FSE 2022a, (New York, NY, USA), p. 344–356, Association for Computing Machinery, 2022.

[7] F. D. Davis, "Perceived usefulness, perceived ease of use, and user acceptance of information technology," *Management Information Systems Quarterly*, vol. 13, pp. 319–340, 09 1989.

[8] D. Marikyan, S. Papagiannidis, and G. Stewart, "Technology acceptance research: Meta-analysis," *Journal of Information Science*, vol. 52, pp. 429–450, 08 2023.

[9] V. Venkatesh, M. G. Morris, G. B. Davis, and F. D. Davis, "User acceptance of information technology: Toward a unified view," *MIS Quarterly*, vol. 27, no. 3, pp. 425–478, 2003.

[10] S. Sternad and S. Bobek, "Tam-based external factors related to erp solutions acceptance in organizations," *International Journal of Information Systems and Project Management*, vol. 1, pp. 25–38, 02 2022.

[11] H. Jo and D.-H. Park, "Mechanisms for successful management of enterprise

resource planning from user information processing and system quality perspective," *Scientific Reports*, vol. 13, no. 1, p. 12678, 2023.

[12] P. Singh, "Usability and productivity for silicon debug software: A case study," Master's thesis, University of Texas at Austin, Austin, TX, USA, Dec 2011.

[13] C.-W. Liu and C.-F. Chien, "An intelligent system for wafer bin map defect diagnosis: An empirical study for semiconductor manufacturing," *Engineering Applications of Artificial Intelligence*, vol. 26, no. 5, pp. 1479–1486, 2013.

[14] P. Lessel, M. Müller, and A. Krüger, "Towards a novel issue tracking system for Industry 4.0 environments," in *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA)*, CHI EA '15, (New York, NY, USA), pp. 1809–1814, Association for Computing Machinery, 2015.

[15] S. Ntoa, "Usability and user experience evaluation in intelligent environments: A review and reappraisal," *International Journal of Human-Computer Interaction*, pp. 1–30, 09 2024.

[16] S. A. C. Perrig, L. F. Aeschbach, N. Scharowski, N. von Felten, K. Opwis, and F. Brühlmann, "Measurement practices in user experience (ux) research: a systematic quantitative literature review," *Frontiers in Computer Science*, vol. Volume 6 - 2024, 2024.

[17] R. M. Innecco, H. C. Martins, and C. A. Gonçalves, "Use of the manufacturing execution system in practice: Understanding the end users and the acceptance of the system," *Revista Gestão da Produção Operações e Sistemas*, vol. 17, p. 57, Jun. 2022.

[18] O. Gelo, D. Braakmann, and G. Benetka, "Quantitative and qualitative research: Beyond the debate," *Integrative Psychological and Behavioral Science*, vol. 42, no. 3, pp. 266–290, 2008.

[19] Illinois Institute of Technology Library, "Quantitative and Qualitative Research." `https://guides.library.iit.edu/c.php?g=1481358&p=11040919`. Accessed: 2026-03-25.

[20] C. Wohlin, "Guidelines for snowballing in systematic literature studies and a replication in software engineering," in *Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering*, EASE '14, (New York, NY, USA), Association for Computing Machinery, 2014.

[21] A. Inan Nur, H. B. Santoso, and P. O. Hadi Putra, "The method and metric of user experience evaluation: A systematic literature review," in *Proceedings of the 2021 10th International Conference on Software and Computer Applications*, ICSCA '21, (New York, NY, USA), p. 307–317, Association for Computing Machinery, 2021.

[22] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative Research in Psychology*, vol. 3, no. 2, pp. 77–101, 2006.

[23] K. Dröge, "noScribe: Ai-powered audio transcription, version 0.6." `https://github.com/kaixxx/noScribe`, 2025. GitHub repository. Accessed: 2026-05-13.

[24] R. Olsen, "Self-selection bias," in *Encyclopedia of Survey Research Methods*, pp. 809–809, Thousand Oaks, California: Sage Publications, Inc., 2008. Accessed: 2026-04-22.

[25] T. Zimmermann, R. Premraj, J. Sillito, and S. Breu, "Improving bug tracking systems," in *2009 31st International Conference on Software Engineering - Companion Volume*, pp. 247–250, 2009.

[26] L.-X. Yin and H.-C. Lin, "Predictors of customers' continuance intention of mobile banking from the perspective of the interactivity theory," *Economic Research-Ekonomska Istraživanja*, vol. 35, no. 1, pp. 6820–6849, 2022.

[27] V. Venkatesh and F. Davis, "A theoretical extension of the technology acceptance model: Four longitudinal field studies," *Management Science*, vol. 46, pp. 186–204, 02 2000.

[28] J. Nielsen, "Response times: The 3 important limits," 1993. Nielsen Norman Group. Accessed: 2026-04-22.

[29] A. Canal Marques, J.-M. Cabrera, and C. de Fraga Malfatti, "Printed circuit boards: A review on the perspective of sustainability," *Journal of Environmental Management*, vol. 131, pp. 298–306, 2013.

Chapter 7

Appendix

7.1 Questions

Performance Expectancy

I would find the system useful in my job.
Using the system enables me to accomplish tasks more quickly.
Using the system increases my productivity.
If I use the system, I will increase my chances of getting a raise.

Do you believe the system is useful for your work? If so, in what way? Do you feel that it affects your productivity? In what way?

Effort Expectancy

My interaction with the system would be clear and understandable.
It would be easy for me to become skillful at using the system.
I would find the system easy to use.
Learning to operate the system is easy for me.

Are there reasons why the system is easy or hard to use? Are there things that can be done to make it easier?

What does the learning curve for the system look like? Are there things that make it easier or harder to learn?

Social Influence

People who influence my behavior think that I should use the system.
People who are important to me think that I should use the system.
Senior management has been helpful in the use of the system.
The organization has supported the use of the system.

How is the system viewed by people around you? Are those who are expected to use it actually using it? Are you encouraged to use the system?

Facilitating Conditions

I have the resources necessary to use the system.
I have the knowledge necessary to use the system.
The system is not compatible with other systems I use.
A specific person or group is available for assistance with system difficulties.

Is it possible to get help and support when using the system? If so, what kind of support is available?



Faculty of Computing, Blekinge Institute of Technology, 371 79 Karlskrona, Sweden